

CONCEPT: Properties of Logarithms

For positive numbers b , m , and n with $b \neq 1$, the following properties hold.

Product Property of
Logarithms

$$\log_b(mn) = \log_b m + \log_b n$$

Quotient Property of
Logarithms

$$\log_b \left(\frac{m}{n} \right) = \log_b m - \log_b n$$

Power Property of Logarithms

$$\log_b(m^n) = n \log_b m$$
